Scalability & Future Plan

Date	15 March 2024
Team ID	SWTID-2026-8204
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Runs only on Flask's single-threaded development server	Cannot handle high concurrent traffic in production	High
2	Model trained on a static, fixed dataset	Recommendations may not reflect newer regional/climate data over time	Medium
3	No database — relies on flat CSV and pickle files	Harder to track prediction history or scale data management	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local Flask dev server, low concurrency	Deploy with Gunicorn/Nginx or migrate to cloud hosting (AWS/Heroku) with load balancing
2	Data Storage	Static CSV dataset + pickle model file	Migrate to a proper database (PostgreSQL/MongoDB) for prediction logs and larger datasets
3	Performance	Lightweight model, fast on small scale	Introduce caching and asynchronous request handling for higher throughput
4	Security	No authentication, minimal input validation	Add input sanitization, rate limiting, and optional user authentication for production use

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Cloud deployment (AWS/Heroku) with production WSGI server	Q3 2026	Improved scalability and public accessibility
Phase 3	Integration of additional ML models (Random Forest, KNN) with model comparison dashboard	Q4 2026	Higher prediction accuracy and flexibility
Phase 4	Mobile-friendly UI and multilingual support for regional farmers	Q1 2027	Wider accessibility and adoption among non-English speaking farmers